GRADE 7 ACCELERATED · UNIT 5 · LESSONS 10–11

Multi-Step Proportion Problems

Independent Set 5.10 **ANSWER KEY**

Name
Date
Period

Six questions, on your own, in silence. Show the step that does the work.

1. A car does 32 miles per gallon and its tank holds 14 gal. How far does it go on a tank?

448 mi

2. That car starts a 1,000-mi trip with a full tank. How many tanks of gas does the trip take?

3 tanks

1,000 ÷ 448 ≈ 2.23 → 3.

3. A recipe for 4 servings uses 1¼ cups of rice. How much rice for 10 servings?

3⅛ cups

k = 2.5; 2.5 × 1¼.

4. 1 km = 0.62 mi. A train does 120 km/h; a bus does 70 mi/h. Which is faster? Show the speeds in one unit.

The train: 120 km/h = 74.4 mi/h > 70 mi/h

120 × 0.62.

5. Juice is mixed orange : mango : water = 3 : 2 : 5. A batch uses 9 cups of orange. How many cups is the whole batch?

30 cups

k = 3; 10 parts × 3.

6. A plan is drawn at 1 in : 5 ft; a room is 2 in by 3 in on it. 1 ft = 0.305 m. Find the room's area in square metres.

About 13.95 m^2

10 ft by 15 ft; 3.05 m by 4.575 m.